TUDOR BERECHET

PORTFOLIO

PORTFOLIO
(deferredreality.com/)

BLOG
(blog.deferredreality.com/)

RESUMÉ

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CONTACT

LFG. Relocation possible.
I live/d in France, Spain, Denmark, Sweden, Romania.
You can contact me at:

TOOLS, SKILLS, BUZZWORDS

ECS,
Unity3D / DOTS, Burst,
UE5, UE4, UE3,
SteamVR, Valve Index, HTC Vive, OculusVR,
Quest/Go/Rift, MS HoloLens, MagicLeap,

Git, Docker, CI/CD,

Agile, Scrum,
Latex,
Figma,
Jira, Trello, Taskworld, Harvest,

Languages

C,
C#,
C++,
Rust,

SPIR-V, CG, HLSL, GLSL,

Python, Blender Python,
Lua, Javascript,

Processing,
php, Matlab,

Technologies

.NET, Azure, WPF, Xamarin,

Vulkan, wgpu, WebGPU, OpenGL, DirectX,
Cuda, Direct Compute / GPGPU,
QT, GLUT,

Pytorch,

TCP/IP, UDP, WebSockets,

Adobe

Photoshop,
Premiere, AfterEffects,

3D

Blender, BPython,
Cinema4D,
Sculptris, zbrush,

EXPERIENCE AND STUDIES

11+ YEARS OF INDUSTRY,
7 YEARS OF UNIVERSITY,
21+ YEARS OF PROGRAMMING

Self-employed (full time) (july 2025 -)

Consulting and developing projects and independent open-source, royalty-free, cross-platform, ECS, threaded, modern but low-cost game and visualization frameworks. [public XR version] ←(https://blog.deferredreality.com/openxr-vulkan-c++-gamedev-boilerplate/)

C++, memory management & ECS, Vulkan render stack, OpenXR single pass render and input stack, custom volumetric BRDF lighting, Blender.

Work (full time) (april 2023 - may 2025)

"Working for Snoop Dogg". (Seniorly building across a few squads at a live French MMO).
←(https://www.youtube.com/channel/UCzv1t7voB-bxMmXLysT4how)

ECS, DOTS, Engine, asset streaming & tools, terraforming & tools, gameplay behaviours & tools.

(Finishing up on-site @ Prediagnose (see below)) (january 2023 - march 2023)

Work (full time) (october 2020 - november 2022)

Senior Graphics Programmer at Cylindo (Chaosgroup (Vray, Enscape)). Automation of 3D / Web / XR pipeline for product visualization, catalogue, design validation. ←(https://www.cylindo.com)

C++, SPIR-V, Python, web technologies (AWS, docker, three.js etc.), streaming, content generation, automation, integration & reusability, (game) engine plugins, post processing & grading etc.

Steam Early Access, VR (indie work, part time) (april 2020 ~ 2021)

VR & Graphics stack developer on Graffiti Bombing. V1 in Steam beta.
A high fidelity VR volumetric Graffiti Simulator with only moderate amounts of too high ambitions.←(store.steampowered.com/app/561440/Graffiti_Bombing/)

Unity3D 2019.2, volumetric data structures & SDF raytracing, recordable lag-proof uninterrupted lines, ECS, VR interactions, networking-agnostic code, data/texture/LoD streaming, DirectX11+ (for now), custom renderers, command buffers.

Work (startup, part time + ↓) (2020, 2022 - 2023)

Software Engineer and R&D for Prediagnose. Building and assembling tech stacks for secure handheld hardware bio-sampler sensor, app, cloud data analysis & automated diagnosing of bacteria, COVID-19, pollutants. ←(https://www.prediagnose.com/)

MS Azure cloud stack, .NET, Xamarin.Forms, UWP, OAuth/Auth0, Shared Access Signatures storage containers, databases, GDPR, PalmSens.

- TED Talk on the biotech ←(https://youtu.be/mzmvHQYRVMc)

PCG Advisor, Branching Sickness (indie work) (summer - winter 2020)

Invited to be a Procedural Content Generation advisor on Branching Sickness on incorporating my published paper. A Halo Flood fan game with a huge team of devs & volunteers. ←(https://twitter.com/SicknessGame)
←(julian.togelius.com/Mark2015Procedural.pdf)

Unity3D -> UE4, PCG, Voxels, Compute, Grammar System Structure, Distance Fields, Additive Noise.

Work (full time, part time + ↑) (february 2017 - september 2020)

VR Game Developer at KhoraVR (games, simulations, healthcare, art, b2b productions, and DK's first VR arcade). ←(khora-vr.com/)

Gameplay, Software, Tools, Graphics Programming. Early adoption for virtually every headset on the market.

Sample project videos:

- VR Raytraced Compute Sculpter ←(youtu.be/p2nTPQErIql)
- Optimized Exposure Therapy Streaming Platform for Oculus Go ←(https://youtu.be/h93L_TTokcc)

Indie Work (↓ - december 2016)

UE4/C++ custom raytracer with prototype GPU physics for a personal/collab project. ←(twitter.com/CarnegieLemon/status/809951848136404993)

Indie Work, VR (september 2016 - ↑)

VR particle sculpting tool with fluid dynamics:

deferredreality.com/?project=particles

Work (full time) (spring 2015 - summer 2016)

Software Developer at UCS3D, Copenhagen. ←(youtube.com/@UtopianCityScape/videos)
Worked on 2 platforms for VR 3D architectural design validation in cityscapes generated from photogrammetry.

App Systems, City PCG pipeline, Custom Lighting Simulation, Point Cloud Data, VR headsets, Gameplay.

Publication: (FDG 2015)

Paper: Procedural Generation of 3D Caves for Games on the GPU. ←(julian.togelius.com/Mark2015Procedural.pdf), (deferredreality.com/?project=gpgpu)
Presented at Foundations of Digital Games 2015. ←(foundationsofdigitalgames.org/)

Voxels, Compute, Grammar System Structure, Distance Fields, Additive Noise.

Work / Internship (full time) (3rd semester of MSc ↓) (2013 - 2013)

Shipped two mobille / tablet titles. DADIU Productions (30 ECTS).
Danish Academy of Digital Interactive Entertainment, Copenhagen.

Masters (2012 - 2014)

MSc. in Game Technology (120 ECTS). IT University of Copenhagen, Denmark. ←(https://en.itu.dk/)

e.g. Game Engines, Game Development, AI, Game Design, Concept Design, Data Mining, PCG, Graphics Programming, and a bunch of jams, VR, and Dungeons & Dragons :)

"Gap year" Masters (2011 - 2012)

A product-oriented break: an Industrial & Product Design MFA (60 ECTS). Umeå Institute of Design, Sweden. (up with the northern lights) ←(www.teknat.umu.se/english/news/.cid273128)
Product Design processes, Concept Design, Storytelling, Maker stuff (arduinos, kinects, exhibit & party pieces).

e.g. Interaction Design, Service Design, Product Design, Color Theory, Visualization Techniques & Storytelling, Ergonomics, Transportation Design, Art History

CPE (2011)

Certificate of Proficiency in English, level C2. Cambridge ESOL Level 3 International.

Got an A (proud - it's the toughest English test suite)

Bachelors (2008 - 2011)

BSc. in Computer Science (180 ECTS). A.I. Cuza University, Iași, Romania. ←(https://www.uaic.ro/)

e.g. Network Programming, Compilers, Assembly, Matlab, OOP, .NET, Databases, Graph / Search Algorithms, Artificial Intelligence, Web Technologies, Information Security, Software Engineering, Computer Graphics, Bioinformatics

Highschool (2005 - 2008)

C and ActionScript

Events, Camps, Misc (2016 - 2020 - Covid - :)

Apart from various VR/XR, product & tech, games, events (AWE, Unite, Gamescom), in my off time I learn, deconstruct, and ramble about futures design, sustainable things that help, last, and **can't** do evil. (tech + game theory + ethics, social innovation & infrastructure, and perhaps the Aya Miyaguchi kind of zkproofs, quadratic voting and anti corruption tech).

I take part in bootcamps, round table discussions, events and meetups e.g. in:

- Akasha Hub Barcelona & Bucharest
- Speculative Futures←(https://www.futures.design/)
- Founders of Tomorrow (2019) alumni (Impact & Ideation camp (SDG), nonprofit based on the Danish Innovation Institute). ←(foundersoftomorrow.com/) ←(un.org/sustainabledevelopment/sustainable-development-goals/) ←(industriensfond.dk/english/about-the-foundation)
- Tech Festival←(techfestival.co/)
- TechBBQ←(https://techbbq.dk/)
- DareDisrupt←(https://daredisrupt.com/)

I also love ♥ the Dungeons & Dragons scene, with the space's exploratory improv qualities. I've used it for team building, self analysis, and just like to dig it up within the professional world.

⤓ Download as PDF (printable)
←(deferredreality.com/boxdrop/TudorBerechetCVSection_Printable-from_personal_website_-_www.deferredreality.com.pdf)

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